

College of Engineering Kallooppa
Pathanamthitta, Kerala 689603

Department of Computer Science and Engineering
(Cyber Security)



Project Diary

**Project Title: Virtualized Hardware-Style Next Generation Firewall
with AI-Assisted Intrusion Detection and DPI**

Sl. No.	Class Roll No.	University Roll No.	Name of student
1	24	PTA22CC024	ASWIN MANOJ
2	54	PTA22CC054	SABARI S
3	38	PTA22CC038	HISHAM FAIZAL
4	5	PTA22CC005	ACHALA A S

Guide's Name : MR.RAJKUMAR T

Group no : 6

Progress of the Project

Name of student:

University Roll No.

Sl. No.	Project Activity	Description	Date of Completion	Sign of Guide
1.	Searched for topics	Researched about topics related to firewalls in IEEE papers & social medias	17/07/2025	
2.	Found out firewall based papers	Found out many interesting firewall based papers	20/07/2025	
3.	Selected a topic for the project	We found out an interesting topic for the project a virtualized next-gen firewall.	22/07/2025	
4.	Researched about the project topic we selected	Researched on the topic virtualized next-gen firewall and gathered basic information of the project from similar research papers	28/07/2025	
5.	Gathered more information of the project	Gather more details about the basic components of the project like DPI, SSL/TLS, Machine Learning Models, Packet Filtering, Datasets used, etc.	31/07/2025	
6.	PPT for zeroth presentation	Created a PPT for zeroth presentation including problem statement, proposed solution, system architecture, etc.	04/08/2025	
7.	Zeroth Presentation	successfully completed zeroth presentation	07/08/2025	
8.	Further research on the topic	we have done further research on the topic	11/08/2025	
9.	Research on Honeypot	Conducted Research on Honeypot, how it works and how we can implement it in our project.	15/08/2025	
10.	Research on Ransomware Detection	Conducted Research on Ransomware Detection, how it works and how we can implement it in our project	21/08/2025	
11.	Preparation for Literature Review	Conducted research again on research papers for literature review	30/08/2025	
12.	PPT for First Presentation	PPT making for first presentation	12/09/2025	
13.	First Presentation	Successfully completed First Presentation	15/09/2025	

Project Review details

Zeroth Presentation

Date: 07/08/2025		Time: From: 01:00 To: 04:00		
Attendance:				
Sl.No.	Roll No	Name of Student	Signature of student	Signature of Coordinator
1.	PTA22CC005	Achala A S		
2.	PTA22CC024	Aswin Manoj		
3.	PTA22CC038	Hisham Faizal E K		
4.	PTA22CC054	Sabari S		
Work Done by students: ● Gathered information about DPI by testing them ● SSL/TLS connection is tested properly and understood the importance of certificates in websites and how SSL/TLS connection works ● Understood Machine Learning Model to know how it is applicable in the project ● Found out many datasets that can be used for training the machine learning model ● Found a way to generate csv report for review and analyze the threats whether it needed for later patches on the firewall				
Comments:				
Sl.	Name	Remarks	Signature	
1	Project Coordinator			
2	HOD			

Project Review details

First Presentation

Date: 15/09/2025		Time: From: <u>09:00</u> To: <u>12:10</u>		
Attendance:				
Sl.No.	Roll No	Name of Student	Signature of student	Signature of Coordinator
1.	PTA22CC005	Achala A S		
2.	PTA22CC024	Aswin Manoj		
3.	PTA22CC038	Hisham Faizal E K		
4.	PTA22CC054	Sabari S		
Work Done by students: ● Adding additional features to the virtualized NGFW to stand out from existing systems ● Adding a feature ransomware detection & Response ● Brought honeypot feature to the project ● Adaptive Traffic Control is brought ● Brought multi-source threat intel for broader coverage ● Found out how to integrate the whole components of the NGFW				
Comments:				
Sl.	Name	Remarks	Signature	
1	Project Coordinator			
2	HOD			